

fong-spivak-seven-sketches — formula report

4029 inline formulas (section omitted; 1 did not render), 264 display equations, 63 tables, 503 unrecovered image regions.

Unresolved (1)

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
fong-spivak-seven-sketches_ F03716		—	$\left[f ; h ; \iota_{\mathrm{E}}, g \circ k ; \iota_{\mathrm{F}}\right. \\ \left.\right]=(f+g) \boxtimes \textcircled{\text{h}} \textcircled{\text{h}} - [\mathbf{U+27C28}](h+k)$	(not rendered)	—
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value					
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Flagged, not acted on — 3 of 78 shown

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
fong-spivak-seven-sketches-E00054	095	0.526	$\mathbf{3}:=\operatorname{Free}\left(\stackrel{\textstyle v_1}{\bullet}\stackrel{\textstyle f_1}{\longrightarrow}\stackrel{\textstyle v_2}{\bullet}\stackrel{\textstyle f_2}{\longrightarrow}\stackrel{\textstyle v_3}{\bullet}\right)$	$\mathbf{3} := \text{Free} \left(\overset{v_1}{\bullet} \xrightarrow{f_1} \overset{v_2}{\bullet} \xrightarrow{f_2} \overset{v_3}{\bullet} \right)$	$\mathbf{3} := \text{Free} \left(\boxed{\overset{v_1}{\bullet} \xrightarrow{f_1} \overset{v_2}{\bullet} \xrightarrow{f_2} \overset{v_3}{\bullet}} \right)$
fong-spivak-seven-sketches-E00064	112	0.147	$\mathbf{G\,r}:=\stackrel{\textstyle \text{Arrow}}{\longrightarrow}\stackrel{\textstyle \text{source}}{\Rightarrow}\stackrel{\textstyle \text{Vertex}}{\text{no equations}}$	$\mathbf{Gr} := \overset{\text{Arrow}}{\longrightarrow} \overset{\text{source}}{\Rightarrow} \overset{\text{Vertex}}{\text{no equations}}$	$\mathbf{Gr} := \boxed{\overset{\text{Arrow}}{\bullet} \overset{\text{source}}{\xRightarrow{\hspace{1cm}}} \overset{\text{Vertex}}{\bullet} \text{no equations}}$ $\mathbf{DDS} := \boxed{\overset{\text{State}}{\bullet} \overset{\text{next}}{\curvearrowright} \text{no equations}}$
fong-spivak-seven-sketches-E00164	257	0.917	$\stackrel{\textstyle A}{\longrightarrow}\stackrel{\textstyle B}{\bullet}\stackrel{\textstyle C}{\bullet}$	$\overset{A}{\longrightarrow} \overset{B}{\bullet} \quad \overset{C}{\bullet}$	$\boxed{\overset{A}{\bullet} \longrightarrow \overset{B}{\bullet} \quad \overset{C}{\bullet}} \subseteq \left \overset{A}{\bullet} \overset{f}{\xrightarrow{\hspace{1cm}}} \overset{B}{\bullet} \overset{h}{\longrightarrow} \overset{C}{\bullet} \overset{i}{\longrightarrow} \overset{D}{\bullet} \right.$
Conf. MathPix confidence: 0.9 0.5-0.9 <0.5 — no value Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured					

The remaining 75 flagged rows are stated as a count.

Each differs from its scan by fewer than 20 components, which is the tail of the distribution rather than its bulk: 39 are component (C), 36 weak (W). By MathPix confidence: 40 at ≥0.9, 16 in 0.6–0.9, 19 below 0.6, 0 with none. Out/465 sampled twenty rows drawn from the ≥0.9 band corpus-wide and found eighteen purely typographic differences, one typographic plus a crop overrun, one borderline and *no* unambiguous content errors — so a high-confidence flag here is evidence of a rendering difference, not of a defect. Every row is in report.ink.json.

Doubted but correct (3)

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
fong-spivak-seven-sketches-EQ0133	214	■ 0.025	$\begin{aligned} \mu_X &:= \left(X + X \xrightarrow{\mathrm{id}_X, \mathrm{id}_X} X \xleftrightarrow{\mathrm{id}_X} X \right) \\ \eta_X &:= \left(\emptyset \xrightarrow{!_X} X \xleftarrow{\mathrm{id}_X} X \right) \\ \delta_X &:= \left(X \xrightarrow{\mathrm{id}_X} X \xleftarrow{[\mathrm{id}_X, \mathrm{id}_X]} X + X \right) \\ \epsilon_X &:= \left(X \xrightarrow{\mathrm{id}_X} X \xleftarrow{!_X} \emptyset \right) \end{aligned}$	$\mu_X := \left(X + X \xrightarrow{[\mathrm{id}_X, \mathrm{id}_X]} X \xleftrightarrow{id}_X X \right)$ $\eta_X := \left(\emptyset \xrightarrow{!_X} X \xleftarrow{\mathrm{id}_X} X \right)$ $\delta_X := \left(X \xrightarrow{\mathrm{id}_X} X \xleftarrow{[\mathrm{id}_X, \mathrm{id}_X]} X + X \right)$ $\epsilon_X := \left(X \xrightarrow{\mathrm{id}_X} X \xleftarrow{!_X} \emptyset \right)$	$\begin{aligned} \mu_X &:= (X + X \xrightarrow{[\mathrm{id}_X, \mathrm{id}_X]} X \xleftarrow{\mathrm{id}_X} X) \\ \eta_X &:= (\emptyset \xrightarrow{!_X} X \xleftarrow{\mathrm{id}_X} X) \\ \delta_X &:= (X \xrightarrow{\mathrm{id}_X} X \xleftarrow{[\mathrm{id}_X, \mathrm{id}_X]} X + X) \\ \epsilon_X &:= (X \xrightarrow{\mathrm{id}_X} X \xleftarrow{!_X} \emptyset) \end{aligned}$
fong-spivak-seven-sketches-EQ0186	279	■ 0.050	$g!(\mathrm{c}_1) = \{ \}^{12} 3 \{ \}^4$	$g!(c_1) = {}^{12}3^4$	$g!(c_1) = \begin{array}{c} 12 \quad 3 \quad 4 \\ \swarrow \downarrow \searrow \\ \bullet \end{array}$
fong-spivak-seven-sketches-EQ0188	279	■ 0.038	$g_!(\mathrm{c}_3) = \{ \}^{12} \{ \}_\downarrow \{ \}^3 \bullet$	$g!(c_3) = {}^{12} \downarrow^3 \bullet^4$	$g!(c_3) = \begin{array}{c} 12 \quad 3 \quad 4 \\ \swarrow \downarrow \searrow \downarrow \\ \bullet \quad \bullet \end{array}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					